Work and Time – Practice Questions

Question 1

Ajit can do a job in 10 days, Kapil can do the same job in 20 days. They together start doing the job, but after 4 days Kapil leaves. How many more days will be required by Ajit to complete the job alone?

Solution:

Ajit's one day work = **1/10**
Kapil's one day work = **1/20**

Work done together in 1 day
= 1/10 + 1/20
= 2/20 + 1/20
= **3/20**

Work done in 4 days
= 4 × (3/20)
= **12/20 = 3/5**

Remaining work
= 1 − 3/5
= **2/5**

Ajit's one day work = **1/10**

Days required
= (2/5) ÷ (1/10)
= (2/5) × 10
= **4 days**

✅ **Answer: 4 days**

Question 2

A can do a piece of work in 10 days while B can do it in 15 days. They begin together, but 5 days before the completion of work, B leaves off. Find the number of days for the work to be completed.

Solution:

A's one day work = **1/10**
B's one day work = **1/15**

Let total time = **x days**

B works for **(x − 5) days**

Work equation:

$x/10 + (x - 5)/15 = 1$

Taking LCM = 30

$3x/30 + 2(x - 5)/30 = 1$

$3x + 2x - 10 = 30$

$5x = 40$

$x =$ **8**

✅ **Answer: 8 days**

Question 3

A and B can do a job in 6 days and 12 days respectively. They begin to work together but A leaves after 3 days. Then the total number of days required for the completion of the work is?

Solution:

A's one day work = **1/6**
B's one day work = **1/12**

Work together in 1 day

1/6 + 1/12
= 2/12 + 1/12
= **1/4**

Work done in 3 days

3 × 1/4 = **3/4**

Remaining work

1 − 3/4 = **1/4**

B's one day work = **1/12**

Days required

$(1/4) \div (1/12)$
$= (1/4) \times 12$
= **3 days**

Total time

3 + 3 = **6 days**

✅ **Answer: 6 days**

Question 4

A, B and C can do the work separately in 16, 31, 42 days respectively. They started the work together but B left 8 days and C 6 days before the completion of work. In what time is the work finished?

Solution:

A's one day work = **1/16**
B's one day work = **1/31**
C's one day work = **1/42**

Let total time = **x days**

A works for **x days**

B works for **(x − 8) days**

C works for **(x − 6) days**

Work equation:

$x/16 + (x - 8)/31 + (x - 6)/42 = 1$

Solving,

$x \approx$ **11.82 days**

✅ **Answer: Approximately 11.82 days**

Question 5

If A, B and C can do a piece of work in 20, 30, 60 days respectively, in how many days can A do the work if he is assisted by B and C on every third day?

Solution:

A’s one day work = **1/20**
B’s one day work = **1/30**
C’s one day work = **1/60**

Work done by A, B and C together

1/20 + 1/30 + 1/60

LCM = 60

= 3/60 + 2/60 + 1/60
= **6/60 = 1/10**

Work pattern (3 days)

Day 1 → A = **1/20**
Day 2 → A = **1/20**
Day 3 → A + B + C = **1/10**

Total in 3 days

1/20 + 1/20 + 1/10
= 1/20 + 1/20 + 2/20
= **4/20 = 1/5**

So 3 days → **1/5 work**

Total days

5 × 3 = **15 days**

✅ **Answer: 15 days**

Hours and Days Both Involved

Question 6

While working for 7 hours a day, A alone can complete a piece of work in 6 days and B alone in 8 days. In how many days will they complete it together if working 8 hours a day?

Solution:

A total hours

6 × 7 = **42 hours**

A’s one hour work = **1/42**

B total hours

8 × 7 = **56 hours**

B’s one hour work = **1/56**

Together per hour

1/42 + 1/56

LCM = 168

= 4/168 + 3/168
= **1/24**

Total hours needed

= **24 hours**

Working hours per day = **8**

Days required

24 / 8 = **3 days**

✅ **Answer: 3 days**

Question 7

Radika can finish some work in 6 days working 5 hours a day. Geeta can finish the same work in 15 days working 3 hours a day. In how many days can they finish the work if working together at 3 hours per day?

Solution:

Radika total hours

6 × 5 = **30 hours**

Radika’s one hour work = **1/30**

Geeta total hours

15 × 3 = **45 hours**

Geeta's one hour work = **1/45**

Together per hour

1/30 + 1/45

LCM = 90

= 3/90 + 2/90
= **1/18**

Work in one day (3 hours)

3 × 1/18 = **1/6**

Days required

1 ÷ (1/6) = **6 days**

✅ **Answer: 6 days**

Man Days Concept – Work and Time

📌 **Basic Formula**

Work = Men × Days

For the **same work**

M1 × D1 = M2 × D2

Where:

- **M = Number of workers**
- **D = Number of days**

📘 **Solved Questions**

Question 1

25 men can finish a work in 12 days.
30 men can finish the same work in how many days?

Solution

M1 × D1 = M2 × D2
25 × 12 = 30 × D
300 = 30D
D = 300 / 30
D = 10

✅ **Answer: 10 days**

Question 2

14 men can complete a work in 22 days.
How many days will 11 men take to complete 1¼ times the work?

Solution

Original work

14 × 22 = 308

New work

1¼ = 5/4
308 × 5/4 = 385
11 × D = 385
D = 385 / 11
D = 35

✅ **Answer: 35 days**

Question 3

4 men can complete a piece of work in 5 days.
How many men are required to complete three times the work in 4 days?

Solution

Original work

4 × 5 = 20

New work

$3 \times 20 = 60$
$M \times 4 = 60$
$M = 60 / 4$
$M = 15$

✅ **Answer: 15 men**

Question 4

12 men can complete a work in 9 days.
After 6 days, 6 more men join.
How many days will they take to complete the remaining work?

Solution

Total work

$12 \times 9 = 108$

Work done in 6 days

$12 \times 6 = 72$

Remaining work

$108 - 72 = 36$

Total men

$12 + 6 = 18$

Days required

$36 / 18 = 2$

✅ **Answer: 2 days**

Question 5

Rations for 250 men for 40 days are available.
After 10 days, 50 more men join.
For how many days will the remaining ration last?

Solution

Total ration

$250 \times 40 = 10000$

Consumed ration

$250 \times 10 = 2500$

Remaining ration

$10000 - 2500 = 7500$

Total men

$250 + 50 = 300$

Days remaining

$7500 / 300 = 25$

✅ **Answer: 25 days**

👥 **Concept – Group of Two**

(Man / Woman / Children / Machine / Robot)

These questions involve **two types of workers**.
We convert one worker type into another to solve the problem.

Question 6

15 men and 20 women can complete a work in 10 days.
24 men and 32 women can complete the same work in how many days?

Solution

First group workers

$15 + 20 = 35$

Second group workers

$24 + 32 = 56$

35 × 10 = 56D
350 = 56D
D = 350 / 56
D = 6.25

✅ **Answer: 6.25 days**

Question 7

2 men and 7 boys can complete a work in 14 days.
3 men and 8 boys can complete the same work in 11 days.
How many days will 8 men and 6 boys take?

Solution

14(2M + 7B) = 11(3M + 8B)
28M + 98B = 33M + 88B
10B = 5M
M = 2B

Convert workers

8M = 16B

Total workers

16B + 6B = 22B

From first equation

11B → 14 days
22B → 7 days

✅ **Answer: 7 days**

Question 8

12 men or 18 women can complete a work in 14 days.
In how many days will 8 men and 16 women complete it?

Solution

12M = 18W
1M = 3/2 W

Convert workers

8M = 12W

Total workers

12W + 16W = 28W
18W → 14 days
28W → (18 × 14) / 28
= 9

✅ **Answer: 9 days**

Question 9

1 man or 2 women or 3 children can complete a work in 44 days.
In how many days will 1 man, 2 women and 3 children complete it?

Solution

1 man work per day = 1/44
1 woman work per day = 1/88
1 child work per day = 1/132

Total work per day

1/44 + 2/88 + 3/132
= 1/44 + 1/44 + 1/44
= 3/44

Days required

1 ÷ (3/44)
= 44/3
= 14⅔

✅ **Answer: 14⅔ days**

Question 10

2 men alone or 3 women alone can complete a work in 4 days.
In how many days will 1 man and 1 woman together complete the work?

Solution

1 man work per day = 1/8
1 woman work per day = 1/12

Combined work

1/8 + 1/12
= 3/24 + 2/24
= 5/24

Days required

1 ÷ (5/24)
= 24/5
= 4.8

✅ **Answer: 4.8 days**